

Raunak Seksaria

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Education

- **International Institute of Information Technology, Hyderabad, India (2023 - Expected 2028)** **CGPA : 9.01**
B.Tech in Computer Science and MS in Computational Natural Sciences by Research
- **St. James' School, Kolkata, India (2008 - 2023)** *ICSE Class 10 : 98% | ISC Class 12 : 97.75%*

Experience

- **Undergraduate Researcher** - Prof. C. Hens, CCNSB, IIIT-H | Kernel PCA, NetworkX, Matplotlib (May 2025 - Ongoing)
 - Co-authoring a paper that clusters 95-country signed networks into blocs, data collected using LLMs, validated on UNGA/GDELT.
 - Found that a bloc's spatial isolation in this embedding predicts weaker outward cooperation ($r=-0.84$) across 1975-2025.
- **Teaching Assistant** - IIIT-H | Proofs, Graph Theory, Group Theory (Aug 2025 - Dec 2025)
 - Conducted 10+ tutorial classes, graded assignments and exams for 'Discrete Structures' course taken by 275+ students.
- **Software Developer Intern** - Product Labs, IIIT-H | Python (asyncio), FFmpeg, Flask (Github) (May 2025 - June 2025)
 - Architected a pipeline that re-broadcasts YouTube livestreams with audio dubbed into a regional language (<10s latency)
 - Engineered a concurrent asyncio pipeline chaining ASR, NMT and TTS models over 5s FFmpeg stream chunks
 - Served the translated stream to browsers via MPEG-DASH with Flask, syncing dubbed audio to video
- **Technical Lead** - Parent Diaries (HopeLog) | MERN, React Native, Whisper APIs, Socket.IO (Github) (Jan 2025 - Apr 2025)
 - Led a team of 4 on the end-to-end development of a cross-platform (iOS/Android) parenting app, tested with Postman.
 - Built a multilingual speech-to-speech AI advisor, and did few-shot milestone extraction from chat transcripts.
 - Implemented Google OAuth + JWT auth, chat systems, email notifications, vaccination/growth tracking, and a parent forum.

Projects

- **Shared File System** | C, Distributed Systems, POSIX threads, TCP sockets, pytest, ASan, CI (Github)
 - Built a metadata name server, replicated storage servers, and a REPL client over a custom line-based TCP protocol.
 - Enabled concurrent multi-user editing through per-sentence write locks with stable IDs and atomic, crash-safe file writes.
 - Implemented heartbeat failure detection, primary-backup replication, and transparent read failover to promoted replicas.
- **Zero-Allocation Bytecode Engine for Hot-Path Signals** | C++20, perf, clang-tidy, ASan/UBSan, CI, Racket (Github)
 - Built a zero-allocation C++ stack VM compiling S-expression signals to bytecode with compile-time lexical addressing.
 - Benchmarked at $\sim 30\times$ a reference tree-walker via rdtsc/perf counters; gated by differential tests, sanitizers, and CI.
- **MiniCrypt: Cryptographic Stack from Scratch** | Python, FastAPI, React (Github)
 - Implemented AES, RSA, ElGamal, HMAC, Miller-Rabin and Oblivious Transfer without any external cryptographic dependencies.
 - Built a FastAPI + React explorer visualising OWF-to-PRP reductions with live attack demos (IND-CPA, MITM).
- **Bitcoin Fraud Detection** | Python, PyTorch Geometric, XGBoost, scikit-learn, pytest (Github)
 - Benchmarked GCN and GraphSAGE against class-weighted XGBoost on a 203k-node Bitcoin transaction graph.
 - Achieved 0.79 illicit-F1 and 1.00 precision@100, quantifying model collapse after a dark-market distribution shift.

Achievements

- **Dean's List** - Top 10% (Semester 2) | Top 15% (Semester 1), among 300+ students at IIIT Hyderabad
- **JEE Advanced** - Rank 4543 | **JEE Mains** - 99 percentile | **WBJEE** - Rank 288

Skills

- **Programming Languages:** Python, C++, C, JavaScript, Java **Web:** React, React Native, Next.js, Flask, FastAPI, Express.js
- **Libraries and Frameworks:** NumPy, Pandas, NetworkX, PyTorch, FFmpeg, libpcap **Databases:** MySQL, SQLite, MongoDB
- **Testing and Tools:** pytest, Valgrind/ASan/UBSan, perf, clang-tidy, Git, Linux, GitHub Actions (CI), Bash, Wireshark, Postman
- **Open-Source Contribution:** MDN Web Docs (Merged PR [37826](#)): Improved the JavaScript closures guide.

Extra-curricular Activities

- **Co-founder** - Katran (Onaya Foundation) (Dec 2020 - Mar 2023)
 - Upcycled 10000m+ waste fabric, distributed 8000+ dresses to marginalised children, uplifted 40+ marginalised artisans
- **Member, Programming Club, IIIT-H, 2024:** Helped organise 4+ competitive programming events.

Relevant Coursework

Data Structures and Algorithms, Design and Analysis of Algorithms, Design and Analysis of Software Systems (OOP Principles), Database Management Systems, Computer Systems Organisation, Operating Systems and Networks, Machine Learning, Linear Algebra, Probability and Random Processes, Principles of Information Security, Principles of Programming Languages.